# Introduction to Convex Optimization and Gradient Descent

**Reading.** Kevin Tian, *Part VI: Continuous Algorithms*, Sections 1–2; and the accompanying October 20 lecture notes. The last section briefly introduces the higher-dimensional viewpoint from Section 5.1.

## 1 Motivation: From Discrete to Continuous Optimization

Many problems we have studied ask us to choose the best object from a set of possible objects. We can write this common form as

$$\min_{x \in \mathcal{X}} f(x) \qquad \text{or} \qquad \max_{x \in \mathcal{X}} f(x).$$

The variable $x$ is the **decision variable**, the set $\mathcal{X}$ is the **feasible set**, and $f$ is the **objective function**.

| **Problem** | **Feasible objects** | **Objective** |
|---|---|---|
| Interval scheduling | Nonoverlapping subsets | Maximize subset size |
| Minimum spanning tree | Spanning trees | Minimize total weight |
| Shortest path | $s$–$t$ paths | Minimize total weight |

In these examples, the choices are discrete. Today, the decision variable can instead be a real number or a vector of real numbers: $x \in \mathcal{X} \subseteq \mathbb{R}^d$. There may be infinitely many feasible choices!

**Example 1. Approximate Continuous Minimization**

**Input:** A function $f : \mathcal{X} \to \mathbb{R}$, a nonempty feasible set $\mathcal{X} \subseteq \mathbb{R}^d$, and an accuracy parameter $\varepsilon > 0$.
**Output:** A point $\widehat{x} \in \mathcal{X}$ satisfying

$$f(\widehat{x}) \le f(x^\star) + \varepsilon, \qquad x^\star \in \arg\min_{x \in \mathcal{X}} f(x).$$

Here we assume that a minimizer $x^\star$ exists.

Our accuracy target concerns the *objective value*, not necessarily the location of the output. An additive error of $\varepsilon$ need not imply $|\widehat{x} - x^\star| \le \varepsilon$.

### 1.1 A Familiar Example: Maximum Flow

A maximum-flow problem already has a continuous formulation. Let $G = (V, E)$ be a directed graph, with source $s$, sink $t$, and capacities $c_e \geq 0$. For this formulation, assume no edges enter $s$ or leave $t$. Let $x_e$ denote the flow on edge $e$, so the decision variable is $x \in \mathbb{R}^{|E|}$.

We want to maximize

$$f(x) = \sum_{e=(s,v)\in E} x_e$$

subject to the constraints

$$\begin{aligned} 0 \leq x_e \leq c_e \qquad & \text{for every } e \in E, \\ \sum_{e=(u,v)\in E} x_e = \sum_{e=(v,w)\in E} x_e \qquad & \text{for every } v \in V \setminus \{s, t\}. \end{aligned}$$

The first constraints enforce capacity bounds; the second enforce flow conservation. The feasible set $\mathcal{X}$ contains every flow satisfying these constraints.

**Question 1.** Suppose there is a nonzero feasible flow $x$. Why are there infinitely many feasible flows, even though the graph is finite?

For every $\lambda \in [0, 1]$, the scaled flow $\lambda x$ is feasible: capacity inequalities and conservation are preserved. Because $x \neq 0$, different values of $\lambda$ give different flows.

### 1.2 Another Motivation: Fitting a Model to Data

Suppose $x \in \mathbb{R}^d$ records the parameters of a prediction rule. We can use a **loss function** $f(x)$ to measure its errors on a collection of training examples, and choose the parameters by minimizing $f(x)$.

There are infinitely many parameter choices, so enumerating all models is not an option. Instead, we will repeatedly improve a current guess using local information about the objective.

**Important distinction.** The loss of a complicated model need not be convex. Our analysis will separate what smoothness alone guarantees from what additionally requires convexity.

# 2 What Information Can an Algorithm Use?

In the **oracle model**, we ask for information about $f$ at points of our choice. A **value query** returns $f(x)$; a **derivative query** returns $f'(x)$ in one dimension. We count queries separately from the cost of computing them. For the gradient-descent theorems, we assume exact derivative queries and exact real arithmetic.

Derivative queries are natural because a derivative describes how the function changes nearby. Value queries can also *approximate* a derivative through

$$f'(x) \approx \frac{f(x+\delta) - f(x)}{\delta}, \qquad \delta > 0.$$

We will make the approximation error precise after defining smoothness.

## 2.1 Without Structure, Queries May Tell Us Nothing

For an unknown point $z \in [0, 1]$, consider

$$f_z(x) = \begin{cases} 0, & x = z, \\ 1, & x \neq z. \end{cases}$$

Even finding $\widehat{x}$ with $f_z(\widehat{x}) \leq \min f_z + 0.99$ requires finding the exact point $z$.

**Why finitely many queries cannot guarantee success.** Consider a deterministic algorithm and answer every query with 1. After its finite query sequence and output $\widehat{x}$, choose $z$ outside all queried points and outside $\{\widehat{x}\}$. Every answer is consistent with $f_z$, but the output has value 1 while the minimum is 0.

**Takeaway:** A bounded domain and bounded function values are not enough. We need assumptions under which queries provide useful information.

**A contrasting example: unimodality.** If $f$ is strictly decreasing up to a minimizer $c$ and strictly increasing afterward, compare its values at

$$u = \ell + \frac{r-\ell}{3}, \qquad v = \ell + \frac{2(r-\ell)}{3}$$

in a known interval $[\ell, r]$ containing $c$. If $f(u) < f(v)$, discard $(v, r]$; if $f(u) > f(v)$, discard $[\ell, u)$; if they are equal, retain $[u, v]$. This is **ternary search**: local queries now reveal which part of the interval can contain the minimizer.

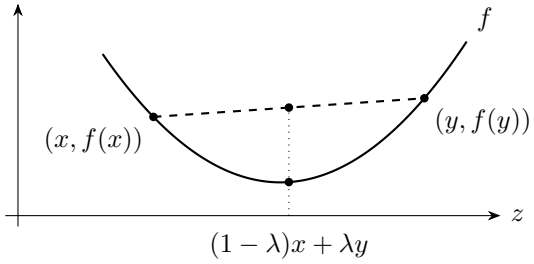
# 3 Convexity: Structure That Helps Optimization

We first work in **one dimension**. Unless otherwise specified, $f : \mathbb{R} \to \mathbb{R}$, and we minimize over all of $\mathbb{R}$.

**Definition 1.** A function $f : \mathbb{R} \to \mathbb{R}$ is **convex** if, for every $x, y \in \mathbb{R}$ and every $\lambda \in [0, 1]$,

$$\boxed{f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y).}$$

The point $(1-\lambda)x + \lambda y$ is a weighted average of $x$ and $y$. The right-hand side is the height of the line segment joining $(x, f(x))$ and $(y, f(y))$. A convex function lies at or below every such segment.




**Question 2.** Which of the following functions are convex on $\mathbb{R}$?

$$f_1(x) = x^2, \quad f_2(x) = |x|, \quad f_3(x) = 3x + 1, \quad f_4(x) = -x^2.$$

**(A)** Only $f_1$.

**(B)** Only $f_1$ and $f_2$.

**(C)** $f_1$, $f_2$, and $f_3$.

**(D)** All four functions.

**Answer: C.** Convex functions need not be differentiable: $|x|$ has a corner at 0. Linear functions satisfy the convexity inequality with equality. For $-x^2$, take $x = -1$, $y = 1$, and $\lambda = 1/2$: the proposed inequality would be $0 \leq -1$.

### 3.1 A Second View: Convex Functions Lie Above Their Tangents

If $f$ is differentiable, its linear, or **Euler**, approximation at $x$ is

$$f(x) + f'(x)(y - x).$$

For a general function, this is only a local approximation. For a convex function, it gives a global lower bound.

**Lemma 1** (First-order inequality)**.** *If $f : \mathbb{R} \to \mathbb{R}$ is differentiable and convex, then, for all $x, y \in \mathbb{R}$,*

$$\boxed{f(y) \geq f(x) + f'(x)(y - x).}$$

*Proof.* For $0 < \lambda \leq 1$, convexity gives

$$f(x + \lambda(y - x)) \leq (1 - \lambda)f(x) + \lambda f(y).$$

Rearrange, dividing by the positive number $\lambda$:

$$f(y) \geq f(x) + \frac{f(x + \lambda(y - x)) - f(x)}{\lambda}.$$

Let $\lambda \to 0^+$. The quotient tends to $f'(x)(y - x)$, which proves the inequality. □

**Two compatible pictures of convexity:**

below every chord and above every tangent.

**Question 3.** Suppose $f$ is differentiable and convex, and $f'(x) = 0$. What can we conclude?

**(A)** $x$ is necessarily the unique minimizer.

**(B)** $x$ is a global minimizer, but need not be unique.

**(C)** $x$ could be a strict local maximum.

**(D)** Nothing about optimality.

**Answer: B.** For every $y$, the first-order inequality yields $f(y) \geq f(x) + 0 \cdot (y - x) = f(x)$. A constant function shows that the minimizer need not be unique.

### 3.2 Why Local Information Is Useful

**Lemma 2** (Local minima are global minima)**.** *For a convex function, every local minimum is a global minimum.*

*Proof.* Suppose $x$ is a local minimum but $f(y) < f(x)$ for some $y$. For sufficiently small $\lambda > 0$, the point $z = (1-\lambda)x + \lambda y$ lies in the neighborhood where $x$ is a minimum. But

$$f(z) \leq (1-\lambda)f(x) + \lambda f(y) < f(x),$$

which is a contradiction. This argument does not require differentiability. □

**The derivative points toward improvement.** Let $x^\star$ be a minimizer. If $f'(x) > 0$, then, for every $y > x$,

$$f(y) \geq f(x) + f'(x)(y-x) > f(x).$$

Thus no minimizer lies to the right of $x$. Similarly, if $f'(x) < 0$, no minimizer lies to the left of $x$. Given an interval containing a minimizer, these tests permit binary search. If $f'(x) = 0$, we have already found a global minimizer.

**Fact 1** (Calculus test)**.** *For twice-differentiable $f : \mathbb{R} \to \mathbb{R}$, convexity is equivalent to $f''(x) \geq 0$ for all $x$. Equivalently, $f'$ is nondecreasing.*

## 4 Gradient Descent: Move Instead of Discard

A one-dimensional interval has a natural left and right. In high dimensions, there is no direct analogue of this interval-based binary search. An alternative is to move our current point using its derivative:

$$\boxed{x_{t+1} = x_t - \eta f'(x_t).}$$

This is **gradient descent**. The parameter $\eta > 0$ is the **step size**.

**The design question:** How should we choose $\eta$? Too small may mean very slow progress; too large may overshoot the minimizer and even increase the objective. Moving in a descent direction is not, by itself, a guarantee that an arbitrary-length step decreases $f$.

# 5 Smoothness: How Far Can We Safely Move?

We need a condition that prevents the derivative from changing too quickly. First, recall a related condition on function values.

**Definition 2.** Let $G \geq 0$ and $L > 0$. A function $f : \mathbb{R} \to \mathbb{R}$ is $G$-**Lipschitz** if, for all $x, y \in \mathbb{R}$,

$$|f(x) - f(y)| \leq G|x - y|.$$

A differentiable function $f : \mathbb{R} \to \mathbb{R}$ is $L$-**smooth** if, for all $x, y \in \mathbb{R}$,

$$\boxed{|f'(x) - f'(y)| \leq L|x - y|.}$$

Lipschitzness of $f$ controls its values; smoothness controls its derivative. We use $G$ and $L$ to distinguish these two bounds.

**Useful calculus tests.** On $\mathbb{R}$, for differentiable $f$, $|f'| \leq G$ everywhere is equivalent to $G$-Lipschitzness. For twice-differentiable $f$, $|f''| \leq L$ everywhere is equivalent to $L$-smoothness.

**Question 4.** Classify each function on *all of* $\mathbb{R}$.

| | **Convex?** | **Lipschitz?** | **Smooth?** |
|---|---|---|---|
| $x^2$ | Yes | No | Yes, $L = 2$ |
| $\lvert x\rvert$ | Yes | Yes, $G = 1$ | No |
| $\cos x$ | No | Yes, $G = 1$ | Yes, $L = 1$ |

**What about simulating derivative queries?** For $L$-smooth $f$ and $\delta > 0$, the fundamental theorem of calculus gives

$$\begin{aligned}\left|\frac{f(x+\delta) - f(x)}{\delta} - f'(x)\right| &\leq \frac{1}{\delta}\int_0^\delta |f'(x+s) - f'(x)|\, ds \\ &\leq \frac{1}{\delta}\int_0^\delta Ls\, ds = \frac{L\delta}{2}.\end{aligned}$$

This is an *approximation*, not an exact simulation. Our theorems use exact derivative queries; finite-difference and floating-point errors are not included in those guarantees.

# 6 Gradient Descent, Take 1: Quadratics

The simplest smooth convex functions are quadratics. Let

$$q(x) = ax^2 + bx + c, \qquad a > 0.$$

We can analyze gradient descent exactly in this case.

**Where is the minimizer?** Complete the square:

$$q(x) = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a},$$
$$x^\star = -\frac{b}{2a}.$$

The additive constant does not affect the location of the minimizer. The error in the objective value is

$$q(x) - q(x^\star) = a(x - x^\star)^2.$$

**What does the derivative tell us?**

$$q'(x) = 2ax + b = 2a(x - x^\star).$$

Thus the derivative is exactly proportional to the signed distance from the current point to the minimizer. Moreover,

$$|q'(x) - q'(y)| = 2a|x - y|,$$

so the tight smoothness constant is $L = 2a$.

**Example 2. Complete the Square** Let $q(x) = x^2 - 6x + 10$. Find the minimizer, the minimum value, the derivative, and a tight smoothness constant.

$$q(x) = (x-3)^2 + 1, \qquad x^\star = 3,$$
$$q(x^\star) = 1, \qquad q'(x) = 2x - 6, \qquad L = 2.$$

**Next question:** Can we choose a step size that lands directly at $x^\star$?

### 6.1 Choosing and Testing the Step Size

For a quadratic with tight smoothness constant $L = 2a$, a gradient step is

$$x_{t+1} = x_t - \eta L(x_t - x^\star).$$

Subtracting $x^\star$ from both sides gives the exact recurrence

$$\boxed{x_{t+1} - x^\star = (1 - \eta L)(x_t - x^\star).}$$

With $\eta = 1/L$, the multiplier is zero: we reach the minimizer in one step. A larger valid upper bound on smoothness gives a smaller, still safe step, but need not give one-step convergence.

**Example 3. Trace the Iterates** Use $q(x) = x^2 - 6x + 10$, $x_0 = 0$, and $L = 2$. Compute $x_1$ and $x_2$ for each choice of $\eta$.

| $\eta$ | $x_1$ | $x_2$ | **Behavior** |
|---|---|---|---|
| 1/4 | 3/2 | 9/4 | Approaches from one side |
| 1/2 | 3 | 3 | Reaches the minimizer |
| 3/4 | 9/2 | 9/4 | Alternates, but gets closer |
| 1 | 6 | 0 | Oscillates without progress |

**Question 5.** Does overshooting always imply that gradient descent diverges? Use the recurrence to determine exactly when the distance decreases for this quadratic.

No. The distance decreases at every nonoptimal iterate precisely when

$$|1 - \eta L| < 1 \quad \Longleftrightarrow \quad 0 < \eta < \frac{2}{L}.$$

For $1/L < \eta < 2/L$, the iterates cross the minimizer but get closer. At $\eta = 2/L$, a nonoptimal starting point oscillates. For $\eta > 2/L$, its distance grows. Thus $1/L$ is a convenient safe choice, not the only convergent choice for a quadratic.

**Progress with $\eta = 1/L$:**

$$q(x) - q\left(x - \frac{q'(x)}{L}\right) = a(x - x^\star)^2 = \frac{(q'(x))^2}{2L}.$$

# 7 Gradient Descent, Take 2: Smooth Functions

A general smooth function is not a quadratic. The useful replacement is that it has a **quadratic upper bound** centered at every point.

**Lemma 3** (Quadratic upper bound)**.** *If $f : \mathbb{R} \to \mathbb{R}$ is $L$-smooth, then, for every $x, y \in \mathbb{R}$,*

$$\boxed{f(y) \le f(x) + f'(x)(y - x) + \frac{L}{2}(y - x)^2.}$$

*Convexity is* not *required.*

**Intuition.** A first-order approximation uses the tangent line. Smoothness controls how far the derivative can move away from its initial value, so the error accumulated along a step is at most quadratic in the step length.

*Proof.* Write $h = y - x$. Parameterizing the segment from $x$ to $y$ by $x + sh$, where $0 \le s \le 1$, gives

$$\begin{aligned} f(y) - f(x) - f'(x)h &= \int_0^1 (f'(x + sh) - f'(x))h \, ds \\ &\le \int_0^1 |f'(x + sh) - f'(x)| \, |h| \, ds \\ &\le \int_0^1 Ls|h|^2 \, ds \\ &= \frac{L}{2}h^2. \end{aligned}$$

The absolute values make the argument valid whether $y \ge x$ or $y < x$. Substitute $h = y - x$ and rearrange. □

For a function that is *both convex and $L$-smooth*, we now have

$$\begin{aligned} f(x) + f'(x)(y - x) &\le f(y) \\ &\le f(x) + f'(x)(y - x) + \frac{L}{2}(y - x)^2. \end{aligned}$$

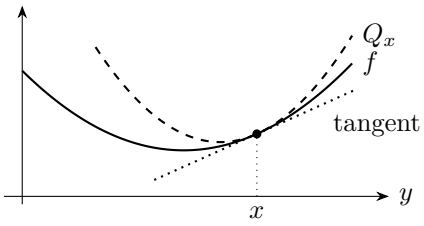
Convexity supplies the linear lower bound.
Smoothness supplies the quadratic upper bound.
These two inequalities play different roles in the analysis.

### 7.1 Minimize the Upper Bound, Not the Unknown Function

At the current point $x$, define the quadratic model

$$Q_x(y) = f(x) + f'(x)(y - x) + \frac{L}{2}(y - x)^2.$$

We know $Q_x(x) = f(x)$ and $f(y) \leq Q_x(y)$ for all $y$.




Differentiate the quadratic model with respect to $y$:

$$Q'_x(y) = f'(x) + L(y - x).$$

Setting this equal to zero gives its minimizer

$$y = x - \frac{f'(x)}{L}.$$

Thus gradient descent with $\eta = 1/L$ minimizes a valid quadratic upper bound at every iteration.

**Lemma 4** (One-step progress)**.** *For any $L$-smooth $f$, the point $x^+ = x - f'(x)/L$ satisfies*

$$\boxed{f(x^+) \leq f(x) - \frac{(f'(x))^2}{2L}.}$$

*Proof.* Substitute $y = x - f'(x)/L$ into Lemma 3:

$$f(x^+) \leq f(x) - \frac{(f'(x))^2}{L} + \frac{L}{2}\frac{(f'(x))^2}{L^2} = f(x) - \frac{(f'(x))^2}{2L}.$$

□

# 8 The Algorithm and Its Two Possible Guarantees

**Algorithm 1** Gradient-Descent$(f, x_0, L, T)$

**Require:** Derivative access to an $L$-smooth function $f : \mathbb{R} \to \mathbb{R}$; $L > 0$; initial point $x_0$; integer $T \geq 1$

$g_{\text{best}} \leftarrow +\infty$
**for** $t = 0, 1, \ldots, T-1$ **do**
  $g_t \leftarrow f'(x_t)$
  **if** $|g_t| < g_{\text{best}}$ **then**
    $\widehat{x} \leftarrow x_t$; $g_{\text{best}} \leftarrow |g_t|$
  **end if**
  $x_{t+1} \leftarrow x_t - g_t/L$
**end for**
**return** $x_T$ and $\widehat{x}$

**Which output?** We use $\widehat{x}$ for a small-derivative guarantee, and $x_T$ for the convex objective-value guarantee. There is no need to store all previous iterates: retaining the current point and the best observed derivative magnitude is enough.

**Runtime.** The algorithm makes $T$ derivative queries. If one derivative costs $C_{\text{grad}}$ time, the total time is $O(T(C_{\text{grad}} + 1))$ in the real-arithmetic model.

**Definition 3.** A point $x$ is an **$\varepsilon$-critical point** if $|f'(x)| \leq \varepsilon$. A point $x$ is an **$\varepsilon$-approximate minimizer** if $f(x) - f(x^\star) \leq \varepsilon$.

These are **different guarantees**. Smoothness alone will let us find an approximate critical point. Convexity and a distance bound will connect criticality to global optimality.

**Question 6.** Let $f(x) = \cos x$. Gradient descent starts at $x_0 = 0$. What happens, and is the output a local minimum?

Since $f'(0) = -\sin 0 = 0$, every iterate remains at 0. But $f(0) = 1$ is a global maximum, while $f(\pi) = -1$. The function is 1-smooth. Thus even an exactly critical point need not be a local minimum of a smooth, nonconvex function.

**Scope reminder.** We are minimizing over $\mathbb{R}$. At the boundary of a constrained feasible set, an optimum need not have derivative zero.

# 9 Analysis I: Finding an Approximate Critical Point

**Theorem 1** (Smooth, possibly nonconvex objective)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be $L$-smooth, with a minimizer $x^\star$, and suppose $\Delta \geq 0$ satisfies*

$$f(x_0) - f(x^\star) \leq \Delta.$$

*After $T \geq 1$ iterations of Algorithm 1,*

$$\boxed{\min_{0 \leq t < T} |f'(x_t)| \leq \sqrt{\frac{2L\Delta}{T}}.}$$

*In particular, $T \geq 2L\Delta/\varepsilon^2$ suffices for the returned $\widehat{x}$ to be an $\varepsilon$-critical point.*

**Proof idea.** A large derivative forces a substantial decrease in $f$. But the total decrease cannot exceed the initial gap $\Delta$.

*Proof.* By Lemma 4, for every $0 \leq t < T$,

$$\frac{(f'(x_t))^2}{2L} \leq f(x_t) - f(x_{t+1}).$$

Sum over the $T$ iterations. The right-hand side telescopes:

$$\begin{aligned} \frac{1}{2L} \sum_{t=0}^{T-1} (f'(x_t))^2 &\leq f(x_0) - f(x_T) \\ &\leq f(x_0) - f(x^\star) \leq \Delta. \end{aligned}$$

The smallest squared derivative is at most their average, so

$$\min_{0 \leq t < T} (f'(x_t))^2 \leq \frac{1}{T} \sum_{t=0}^{T-1} (f'(x_t))^2 \leq \frac{2L\Delta}{T}.$$

Taking square roots proves the result. □

**Question 7.** Holding $L$ and $\Delta$ fixed, how does the sufficient iteration count change if the desired derivative tolerance decreases from $\varepsilon$ to $\varepsilon/2$?

It increases by a factor of 4, since the bound is proportional to $1/\varepsilon^2$. Notice that no part of the proof used convexity.

# 10 Analysis II: Adding Convexity

For a convex objective, Lemma 1 gives

$$\begin{aligned} f(x) - f(x^\star) &\le f'(x)(x - x^\star) \\ &\le |f'(x)|\,|x - x^\star|. \end{aligned}$$

A small derivative therefore gives a small objective gap *provided* we also control the distance to a minimizer.

**Lemma 5** (Distance never increases)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be convex and $L$-smooth, with minimizer $x^\star$. For $x^+ = x - f'(x)/L$,*

$$|x^+ - x^\star| \le |x - x^\star|.$$

*Consequently, if $|x_0 - x^\star| \le R$, then $|x_t - x^\star| \le R$ for every gradient-descent iterate.*

*Proof.* Write $g = f'(x)$. Expanding the squared distance gives

$$(x^+ - x^\star)^2 = (x - x^\star)^2 - \frac{2}{L} g(x - x^\star) + \frac{g^2}{L^2}.$$

We need $g(x - x^\star) \ge g^2/(2L)$. To prove this, chain together three facts:

$$\begin{aligned} g(x - x^\star) &\ge f(x) - f(x^\star) && \text{by convexity} \\ &\ge f(x) - f(x^+) && \text{by optimality of } x^\star \\ &\ge \frac{g^2}{2L} && \text{by one-step progress.} \end{aligned}$$

Substituting yields

$$(x^+ - x^\star)^2 \le (x - x^\star)^2 - \frac{2}{L}\frac{g^2}{2L} + \frac{g^2}{L^2} = (x - x^\star)^2.$$

Both distances are nonnegative, so taking square roots proves the one-step claim. Apply it repeatedly to obtain the statement for every $t$. □

**What to remember:** Smoothness controls how much a step can change $f$; convexity supplies the inequality that also controls the distance.

## 10.1 A First Global Convergence Guarantee

**Theorem 2** (Introductory convex guarantee)**.** *Suppose $f : \mathbb{R} \to \mathbb{R}$ is convex and $L$-smooth, attains its minimum at $x^\star$, and $|x_0 - x^\star| \le R$, where $R > 0$. For every $T \ge 1$, Algorithm 1 satisfies*

$$\boxed{f(x_T) - f(x^\star) \le \frac{LR^2}{\sqrt{T}}.}$$

*Hence $T \ge L^2R^4/\varepsilon^2$ suffices for an $\varepsilon$-approximate minimizer.*

**Proof idea.** Bound the initial objective gap, find a small derivative, and use convexity together with the distance invariant.

*Proof.* **Step 1: Bound the initial gap.** Since $x^\star$ is an unconstrained minimizer of a differentiable function, $f'(x^\star) = 0$. Center Lemma 3 at $x^\star$:

$$f(x_0) - f(x^\star) \le \frac{L}{2}(x_0 - x^\star)^2 \le \frac{LR^2}{2}.$$

Thus we may use $\Delta = LR^2/2$ in Theorem 1.

**Step 2: Find an iterate with a small derivative.** For some $0 \le t < T$,

$$|f'(x_t)| \le \sqrt{\frac{2L\Delta}{T}} = \frac{LR}{\sqrt{T}}.$$

**Step 3: Convert criticality to global optimality.** Convexity and Lemma 5 imply

$$f(x_t) - f(x^\star) \le |f'(x_t)|\,|x_t - x^\star| \le \frac{LR}{\sqrt{T}}\,R = \frac{LR^2}{\sqrt{T}}.$$

Finally, Lemma 4 says the objective values never increase. Since $t < T$, we have $f(x_T) \le f(x_t)$, which proves the claim for the *last* iterate. □

**Equivalent accuracy calculation.** To make the objective gap at most $\varepsilon$, it suffices to find a derivative of magnitude $\varepsilon/R$:

$$\frac{2L(LR^2/2)}{(\varepsilon/R)^2} = \frac{L^2R^4}{\varepsilon^2} \quad \text{iterations suffice.}$$

# 11 The Higher-Dimensional Viewpoint

Now let $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$. A set $\mathcal{X} \subseteq \mathbb{R}^d$ is **convex** if it contains the entire line segment between every pair of its points:

$$x, y \in \mathcal{X},\ \lambda \in [0,1] \quad \Longrightarrow \quad (1-\lambda)x + \lambda y \in \mathcal{X}.$$

A function $f : \mathcal{X} \to \mathbb{R}$ is convex if $\mathcal{X}$ is convex and

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y).$$

Minimizing a convex function over a convex set is a **convex optimization problem**. Our gradient-descent analysis concerns the unconstrained case $\mathcal{X} = \mathbb{R}^d$.

**Replace the derivative with a gradient.** For differentiable $f : \mathbb{R}^d \to \mathbb{R}$, the gradient is

$$\nabla f(x) = \begin{pmatrix} \partial f(x)/\partial x_1 \\ \vdots \\ \partial f(x)/\partial x_d \end{pmatrix}.$$

Write $\langle u, v \rangle = \sum_i u_i v_i$ and $\|u\|_2 = \sqrt{\sum_i u_i^2}$. Euclidean $L$-smoothness means

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2.$$

The update becomes

$$\boxed{x_{t+1} = x_t - \frac{1}{L}\nabla f(x_t).}$$

**Example 4. A Two-Dimensional Step** Let $f(u, v) = u^2 + 4v^2$. Verify that $L = 8$ is valid, and take one gradient descent step from $(2, 1)$ with $\eta = 1/8$.

The gradient is $(2u, 8v)$, and $\|(2a, 8b)\|_2^2 = 4a^2 + 64b^2 \leq 64(a^2 + b^2)$. At $(2, 1)$, the gradient is $(4, 8)$, so the next point is $(2, 1) - (4, 8)/8 = (3/2, 0)$. A single step size need not solve both coordinates in one step.

### 11.1 The Same Analysis, With Inner Products and Norms

The two central inequalities generalize to

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \qquad \text{(convexity)},$$
$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|_2^2 \qquad \text{(smoothness)}.$$

In particular, the same substitution into the upper bound gives

$$f\left(x - \frac{1}{L}\nabla f(x)\right) \leq f(x) - \frac{\|\nabla f(x)\|_2^2}{2L}.$$

The telescoping proof therefore gives

$$\min_{0 \leq t < T} \|\nabla f(x_t)\|_2 \leq \sqrt{\frac{2L\Delta}{T}}.$$

For the convex guarantee, replace squared scalar distances by squared Euclidean distances in Lemma 5, and use Cauchy–Schwarz:

$$f(x) - f(x^\star) \leq \langle \nabla f(x), x - x^\star \rangle \leq \|\nabla f(x)\|_2 \, \|x - x^\star\|_2.$$

Thus the introductory bound in Theorem 2 also holds when $\|x_0 - x^\star\|_2 \leq R$.

**Iterations are not the same as runtime.** There is no explicit $d$ in these iteration bounds. However, $L$, $R$, and the cost of a gradient can depend on $d$. Each vector update also takes $O(d)$ arithmetic operations, so $T$ iterations cost

$$O\big(T(C_{\text{grad}} + d)\big).$$

**Question 8.** Which statement is justified by today's analysis?

**(A)** A small gradient always means a nearby minimizer.

**(B)** Smoothness alone guarantees a global minimum.

**(C)** A constant number of scalar arithmetic operations always suffices for one iteration, regardless of $d$.

**(D)** Smoothness guarantees progress proportional to the squared gradient norm; convexity can turn local information into a global bound.

**Answer: D.** A confuses derivative size with distance to a minimizer; B assumes smoothness implies global optimality; C ignores the cost of processing $d$ coordinates.

## 12 Summary and Beyond Today's Analysis

**The algorithm.** Start from $x_0$ and repeatedly use the local slope to improve the current guess. In one dimension,

$$x_{t+1} = x_t - \frac{1}{L} f'(x_t).$$

In $\mathbb{R}^d$, replace $f'(x_t)$ with $\nabla f(x_t)$.

**The role of each assumption.** **Smoothness** gives a quadratic upper bound and a safe step size. **Convexity** gives a linear lower bound and rules out nonglobal local minima. A bound on the initial gap or distance turns these structural statements into an iteration count.

**The guarantees proved in this lecture.** For a smooth objective with initial value gap at most $\Delta$,

$$T \geq \frac{2L\Delta}{\varepsilon^2} \quad \Longrightarrow \quad \text{an } \varepsilon\text{-critical point among the queried iterates.}$$

For a smooth convex objective with initial distance at most $R$,

$$T \geq \frac{L^2 R^4}{\varepsilon^2} \quad \Longrightarrow \quad f(x_T) - f(x^\star) \leq \varepsilon.$$

Take the ceiling of the displayed iteration bound, and at least one iteration. These results assume an attained optimum and exact gradients.

**Stronger results: stated, not proved here.** The supplied notes, Section 2.4, point out that a sharper analysis of the *same* gradient-descent algorithm gives an $\varepsilon$-approximate minimizer in $O(LR^2/\varepsilon)$ iterations. The $1/\varepsilon^2$ dependence above is a limitation of our introductory proof, not the best bound for gradient descent.

The notes also describe **acceleration**: an appropriately modified algorithm using momentum attains $O(\sqrt{LR^2/\varepsilon})$ iterations for smooth convex minimization. Arbitrarily adding momentum does not inherit this guarantee; the method and its analysis are beyond today's lecture.

**Sources.** Kevin Tian, *CS331, Part VI: Continuous Algorithms*, Sections 1–2 (especially Lemmas 1–5) and Section 5.1; and *CS331, Fall 2025, Lecture 15 (10/20)*. These sources supply the theory and lecture sequence. Exercises and questions were written for this adaptation.